10 Overgeneral autobiographical memories and their relationship to rumination

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Introduction

Overgeneral autobiographical memory (Williams et al., 2007) is the recall of categoric summaries of repeated events (e.g., "playing golf every week") even when asked to recall specific memories that are located at a particular place and time (e.g., "beating my friend Paul at golf last Saturday"). Overgeneral memory recall is clinically important because it is elevated in depressed patients (e.g., Kuyken & Brewin, 1995; Williams & Broadbent, 1986) and formerly depressed patients compared with controls (Brittlebank et al., 1993; Mackinger et al., 2000) and predicts poorer long-term outcomes including future depression (Brittlebank et al., 1993; Dalgleish et al., 2001; Mackinger et al., 2000; for review, see Williams et al., 2007). Thus, categoric autobiographical memory is either a marker for or itself a vulnerability factor for depression.

As research has sought to understand the mechanisms underpinning the development and maintenance of overgeneral autobiographical memory in depression, convergent evidence has increasingly implicated a key role for ruminative thinking. More recently, there is also preliminary evidence that manipulating the specificity of autobiographical memory recall may itself influence rumination. This chapter therefore briefly summarizes the rumination literature and then reviews the evidence that rumination causally influences overgeneral autobiographical memory, and vice versa. Through this chapter, the concept of abstract processing will emerge and be discussed as a unifying construct across rumination and overgeneral memory.

Rumination

Rumination is defined as "behaviours and thoughts that focus one's attention on one's depressive symptoms and on the implications of those symptoms" (Nolen-Hoeksema, 1991, p. 569). In practice, rumination involves repeated dwelling on the self, depressed symptoms, upsetting events, and personal problems, for example, "Why did this happen to me? Why do I feel like this? Why do I always react this way?"

Both prospective longitudinal and experimental studies have confirmed that rumination plays an important causal role in the onset, maintenance, and recurrence of depression. Rumination prospectively predicts the onset of major depressive episodes and the severity of depressive symptoms in nondepressed and currently depressed individuals, and mediates the effects of other risk factors on depression (e.g., Nolen-Hoeksema, 2000; Spasojević & Alloy, 2001; see meta-analysis by Mor & Winquist, 2002; reviews by Nolen-Hoeksema et al., 2008; Watkins, 2008). Rumination is elevated in women relative to men, and in currently depressed and recovered depressed patients relative to healthy controls.

Rumination also prospectively predicts symptoms of anxiety after controlling for baseline anxiety, as shown in numerous longitudinal studies (Watkins, 2008). For example, rumination predicts the onset and severity of posttraumatic stress symptoms and diagnosis of posttraumatic stress disorder (PTSD) up to three years after traumatic events (e.g., Ehlers et al., 1998). Moreover, two large-scale longitudinal studies found that rumination explained the concurrent and prospective associations between symptoms of anxiety and depression (McLaughlin & Nolen-Hoeksema, 2011). Rumination also prospectively predicts substance abuse (Nolen-Hoeksema et al., 2007; Skitch & Abela, 2008), alcohol abuse (Caselli et al., 2010), and eating disorders (Holm-Denoma & Hankin, 2010; Nolen-Hoeksema et al., 2007), after controlling for initial symptoms. These convergent data across studies have led to the proposal that rumination may be a transdiagnostic process, that is, a process present across multiple psychiatric diagnoses, and that causally contributes to those disorders (Ehring & Watkins, 2008; Harvey et al., 2004; Nolen-Hoeksema & Watkins, 2011).

Furthermore, there is extensive experimental evidence that manipulating rumination causally influences existing negative affect and negative cognition, suggesting that it may be a key driver in the activation of maladaptive patterns of information processing associated with an episode of depression. Studies have used a standardized rumination induction, in which participants are instructed to spend eight minutes concentrating on a series of sentences that involve rumination about themselves, their current feelings, and physical state and the causes and consequences of their feelings (e.g., "Think about the way you feel inside"; Lyubomirsky & Nolen-Hoeksema, 1995). As a control condition, a distraction induction is typically used, in which participants are instructed to spend eight minutes concentrating on a series of sentences that involve imagining visual scenes that are unrelated to the self or to current feelings (e.g., "Think about a fire darting round a log in a fireplace").

Compared with the distraction induction, the rumination induction is reliably found to have negative consequences on mood and cognition, but only when participants are already in a negative rather than a neutral mood (e.g., selected dysphoric or depressed participants, or following a sad mood induction). Thus, for participants in a sad mood, compared with distraction, rumination exacerbates negative mood, increases negative thinking about the self, increases negative autobiographical memory recall, increases negative thinking about the future, impairs concentration and central executive functioning, and impairs social problem solving (e.g., Lyubomirsky & Nolen-Hoeksema, 1995; Rimes & Watkins, 2004; Watkins & Brown, 2002; Watkins & Teasdale, 2001; see review in Watkins, 2008). Thus, ruminative thinking can act to fuel a negative spiral between cognition and emotion, resulting in depression.

This convergent evidence indicates the importance of rumination in depression, of understanding the mechanisms underpinning rumination, and of deriving tractable theoretical accounts of rumination. Experimental attempts to disentangle the mechanisms of rumination further revealed that there are distinct modes of processing within rumination, each with distinct functional effects (adaptive versus maladaptive, e.g., Watkins & Moulds, 2005; Watkins & Teasdale, 2001; 2004). These studies led to the elaboration of a processing mode theory, which distinguishes between abstract versus concrete processing modes within rumination (Watkins, 2008). An abstract processing mode involves focusing on general, superordinate, and decontextualized mental representations that convey the essential meaning, causes, and implications of goals and events, including the "why" aspects of an action and the ends consequential to it. Such abstract thinking is characteristic of the phenomenology of depressive rumination. In contrast, a concrete processing mode involves a focus on the direct, specific, and contextualized experience of an event and on the details of goals, events, and actions that denote the feasibility, mechanics, and means of "how" to do the action. The processing mode theory hypothesized that abstract processing of negative information is maladaptive relative to concrete processing. The rationale behind the processing mode theory and relevant evidence is described in more detail in the section titled "Abstract Processing Causally Influences Rumination and Its Consequences."

Rumination causally influences overgeneral autobiographical memory

When examined in patients with depression, the standard experimental manipulations of rumination versus distraction were found to influence overgeneral autobiographical memory recall: compared with distraction, rumination reduced the specificity of autobiographical memory retrieval (Kao et al., 2006; Park et al., 2004; Watkins & Teasdale, 2001; 2004; Watkins et al., 2000; see Williams et al., 2007 for discussion).

Watkins and Teasdale (2001) further distinguished between two components within the standard conceptualization of rumination: focus on the self and symptoms versus a particular processing style or mode characterized by analytical-evaluative thinking focused on meanings and implications (e.g., asking "Why me?"). To investigate hypothesized differential effects of self-focus versus the analytical processing mode, Watkins and Teasdale (2001) developed four alternative versions of the standard experimental manipulations to manipulate these different dimensions in a 2 × 2 design: high analytical, high self-focus (rumination); low analytical, low self-focus (distraction); low analytical, high self-focus (experiential self-focus); high analytical, low self-focus (abstraction). Thus, the two high self-focus manipulations were equivalent in focus on mood, symptoms, and self but differ in the mode of thinking, either encouraging an analytical, conceptual, abstract orientation (i.e., starting with the prompt ""Think about the causes, meanings and consequences of…") or an experiential orientation (starting with the prompt "Focus your attention on your experience of…"), each applied to the same self-related and feeling-focused items (e.g., "how happy or sad you feel," "how alert or tired you feel"). The abstraction condition involved focus on abstract non-self-related concepts, such as "Think about whether the ends justify the means." The distraction condition was the same as the previous distraction condition used by Lyubomirsky and Nolen-Hoeksema (1995).

This experiment found a double dissociation between the effects of self-focus and processing mode in patients with depression. First, self-focus but not processing mode influenced self-reported mood in depressed patients following the manipulations: sad mood increased in high self-focus conditions (rumination; experiential self-focus) relative to low self-focus conditions (distraction; abstraction). Second, processing mode but not self-focus influenced the proportion of overgeneral autobiographical memories recalled from before to after the manipulation: the proportion of categoric memories recalled increased in the high analytical/abstract conditions (rumination; abstraction) relative to the low analytical/abstract conditions (experiential self-focus; distraction).

Thus, the mode of processing adopted during rumination (analytical and abstract versus concrete and experiential) influenced the recall of overgeneral autobiographical memories, indicating that this recall was influenced by thinking style at retrieval. This pattern of findings was replicated by Watkins and Teasdale (2004), who compared analytical versus experiential self-focus in depressed patients. Again, these two manipulations had distinct functional effects on overgeneral autobiographical memory recall in depressed patients: conceptual-evaluative self-focus maintained overgeneral autobiographical memory recall, whereas experiential self-focus reduced overgeneral autobiographical memory recall (Watkins & Teasdale, 2004).

These results suggested that rumination can have a causal role in influencing overgeneral autobiographical memory. More specifically, they indicated that reducing rumination is associated with reductions in categoric memory recall and that the abstract processing element of rumination is important. However, this left open the question of whether overgeneral autobiographical memory can be experimentally induced in nondysphoric participants, that is, whether this effect generalizes to other individuals. Moreover, to date, studies had shown only that reducing rumination was associated with reductions in overgeneral autobiographical memory. Definitive evidence of the causal role of rumination in the recall of overgeneral autobiographical memories requires the demonstration that overgeneral memory recall can be induced in a sample that does not typically display it. To test this direction of causality, Barnard et al. (2006) induced setting conditions for rumination experimentally in a nonclinical sample. Phenomenologically, rumination involves perseverative thinking on repetitive and narrow themes about self-related material (typically on negative content). Theoretically, this can be conceptualized as the combination of an abstract processing mode focused on meanings and implications coupled with a relatively impoverished schematic model, such that there is limited scope for a wide range of thoughts, resulting in the same negative thoughts recurring again and again (Barnard et al., 2006).

To recreate this pattern of processing experimentally, Barnard et al. (2006) manipulated a category fluency task. In the "blocked" condition, participants repeatedly generated self-related information on the same theme across each block in the category fluency task to parallel perseverative thinking associated with an impoverished schematic model. In the contrasting "intermixed" condition, participants generated self-related information in the category fluency task within blocks that intermixed prompts across superordinate themes to minimize the recreation of perseverative thinking. Thus, while both conditions generated self-related information in the category fluency task to the same sets of stimuli arranged across blocks, in the "blocked" condition, self-related information was generated on the same theme repeatedly, whereas in the "intermixed" condition, self-related information moved rapidly across themes. As predicted, there were fewer specific autobiographical memories and more categoric memories generated in the blocked (ruminative) condition than in the intermixed condition, but only when the themes were self-related. No difference was found in a subsequent study when the themes related to the activities of animals. Thus, this study provides convergent evidence in a different population and with a different rumination induction that rumination causally contributes to the reduction in specific autobiographical memory recall.

In a further experimental study, Raes et al. (2008) examined whether inducing an abstract evaluative ruminative mode relative to a more concrete, process-focused mode could influence overgeneral memory recall in students. Previous studies had shown that participants trained to think in an abstract or concrete way, through repeated practice at either abstractly evaluating the causes, meanings, and implications of emotional scenarios or imagining the concrete details of how an event is occurring, responded differently to an unanticipated failure. Individuals trained to think about emotional events in a concrete way had reduced emotional reactivity to a subsequent experimental stressor relative to those who were trained to think abstractly (Moberly & Watkins, 2006; Watkins et al., 2008). Raes and colleagues adapted these approaches to examine whether training participants to adopt a more abstract mode would maintain overgeneral autobiographical memory recall in 195 undergraduates. Because the standard Autobiographical Memory Test utilized to assess overgenerality of memory is not sensitive in nonclinical populations, a more sensitive test, the Sentence Completion for Events from the Past Test (SCEPT), was used. This involves participants being provided with a written list of sentence stems (e.g., "I often…"), which they then complete. These completions are then rated on the standard autobiographical memory task categories for whether the completions reflect specific or categoric autobiographical memories.

Participants were exposed to training in the two different modes across twelve emotional scenarios, of which six were positive and six were negative (e.g., "car accident," "birthday surprise"), to match their effects on emotional valence. As predicted, the proportion of overgeneral memories recalled was significantly greater for participants who practiced in the abstract condition relative to participants who practiced in the concrete condition. Furthermore, the proportion of specific autobiographical memories recalled was significantly less for the abstract condition than the concrete condition (for more information on abstract and concrete self-images, see Rathbone and Moulin, Chapter 14).

In summary, multiple studies, with different methodologies, in clinical and nonclinical samples have found that manipulating rumination causally influences the specificity/generality of autobiographical memory recall. These results provide robust evidence consistent with the hypothesis that abstract rumination plays a causal role in the generation of overgeneral autobiographical memory. It is interesting to note, however, that self-reported trait rumination is not always found to be associated with level of memory specificity (see Smets et al., 2013, for evidence of null findings, although with the caveat that these studies were mainly in nonclinical populations). It may be that the relationship between rumination and overgeneral memory is dependent on increased clinical symptoms and/or activation of current state rumination.

These findings in turn informed the Capture and Rumination, Functional Avoidance, and Executive (CaRFAX) model of autobiographical memory (Williams et al., 2007). The CaRFAX model proposes three interacting mechanisms to account for overgeneral memory recall. First, information capture and rumination processes are hypothesized to contribute to the maintenance of overgeneral autobiographical memory retrieval. Williams et al. (2007) argued that the act of starting to retrieve autobiographical memories can activate ruminative thinking because such recall involves accessing information about the self and about personally relevant goals, which is likely to trigger rumination in those with negative self-image and/or unresolved personal goals (Watkins, 2008). Furthermore, because depressive rumination is characterized by abstract processing focused on evaluating the meanings and implications of events, it is likely to result in more abstract and generic mental representations, including the search terms used in autobiographical memory recall.

The CaRFAX model adopts the theoretical framework proposed by Conway and Pleydell-Pearce (2000) to explain the process of autobiographical memory retrieval. Key to this model is the assumption that autobiographical memory retrieval occurs in a generative way when deliberately recalling a memory in response to a request or cue, working through a hierarchy of stages of memory search. In response to a search attempt, general lifetime period information (e.g., “when I was at school”) is interrogated first, which then informs the access to and search for categoric information (e.g., “when waiting for the bus”), which in turn can help to search for relevant specific perceptual and sensory detail, which needs to be added to generate a specific episodic autobiographical memory. Within this framework, the abstract processing that occurs during rumination would focus representations toward the categoric generic stage of the hierarchy of information retrieval and, thus, could interfere with moving toward the construction of a specific memory. Furthermore, the repeated elaboration of such generic descriptions, which would occur during rumination, would make this level of the hierarchy more easily accessible and build up a conceptual network of associated generic descriptions, such that when an individual sought to retrieve specific information with a generic search description, it may lead instead to activation of other generic descriptions, making it harder to access specific information. This process, called "mnemonic interlock," has been proposed as a central mechanism accounting for overgeneral autobiographical memory recall (Williams et al., 2007).

The CaRFAX model also proposes that functional avoidance may be a mechanism underpinning overgeneral autobiographical memory. There is some evidence that specific memories of negative events involve more intense affect. Thus, truncating the memory search at the more generic and categoric stage may be negatively reinforced by avoiding the more intense affect associated with specific memories (for further details, see Van den Broeck and colleagues, Chapter 11).

Finally, the CaRFAX model hypothesizes that impairments in executive capacity and control can underpin reduced specificity of autobiographical memory recall, by reducing the resources necessary for generative retrieval. Moving through the information hierarchy will require executive capacity and cognitive control to inhibit unwanted descriptions, to reduce unwanted emotional responses, and to focus on desired information. However, individuals with impaired executive control, whether due to a persistent individual difference or the consequence of state effects (e.g., depression, fatigue, alcohol) may find it more difficult to efficiently navigate the different stages of memory retrieval, causing them to be stuck at the generic level.

In sum, the CaRFAX model predicts that "overgeneral memory should shift in response to interventions that reduce the accessibility of negative self-schemas, break ruminative processing cycles, and/or train individuals in executive control of cognitive processing" (Williams et al., 2007, p. 143).

Rumination causally influences overgeneralization

In parallel to the evidence that rumination causally contributes to overgeneral autobiographical memory recall, there is evidence that implicates rumination in the generation of other forms of overgeneral thinking, such as overgeneralization. Overgeneralization has been defined as drawing a

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general rule or conclusion on the basis of an isolated incident that is applied across the board to related and unrelated situations. For example, a single negative event, such as doing poorly on a test, is interpreted as indicating a global, characterological inadequacy, such as being stupid. Overgeneralization is considered an important mental process because it is elevated in depression (Beck, 1976; Carver & Ganellen, 1983), appears to be specific to depression and not anxiety (Carver & Ganellen, 1983; Carver et al., 1988; Ganellen, 1988), and prospectively predicts subsequent depression (Carver, 1998; Dykman, 1996; Edelman et al., 1994), paralleling the observed properties of overgeneral autobiographical memory.

Experimental studies have shown that manipulating the processing mode during rumination influences overgeneralization. Rimes and Watkins (2005) found that the abstract-analytical rumination condition increased global self-devaluations, in the form of ratings of worthlessness and competence, relative to the experiential-concrete rumination condition, in depressed patients. Watkins and Moberly (in preparation) compared the effects of repeated training to adopt abstract versus concrete processing modes on overgeneralization in an undergraduate sample. Overgeneralization was assessed using a paradigm developed by Wenzlaff and Grozier (1988), in which each participant rates him- or herself on general trait adjectives (e.g., competence, friendliness) before and after an anagram stressor task described as an intelligence test. The extent to which failure on the task leads to increased endorsement of general trait adjectives unrelated to intelligence provides an index of overgeneralization in response to poor performance on the anagram task. As predicted, the abstract condition resulted in greater overgeneralization (reduced ratings of proficiency and friendliness) following the failure than did the concrete condition. These findings provide further evidence consistent with the processing mode theory and suggest that adoption of an abstract processing mode could account for the elevated tendency for overgeneralization found in depressed and low-self-esteem individuals.

Abstract processing causally influences rumination and its consequences

As noted earlier, Watkins (2008) proposed that the distinct consequences of rumination may depend on the processing mode (concrete versus abstract) adopted during ruminative thinking about self-related information. This theory was initially informed by the findings described above in which different variants of rumination manipulations had differential effects on mood versus specificity of autobiographical memory retrieval (Watkins & Teasdale, 2001; 2004) and was supported by further experimental studies (see below).

The processing mode theory proposes that the consequences of abstract versus concrete processing during rumination are determined by their relative sensitivity to contextual and situational detail. Relative to a concrete mode, an abstract mode (1) insulates an individual from the specific context, (2) makes the individual less distractible and less impulsive, (3) enables more consistency and stability of goal pursuit across time, and (4) allows gainful and unhelpful generalizations and inferences across different situations. However, it also (1) makes the individual less responsive to the environment and to any situational change and (2) provides fewer specific and contextual guides to action and problem solving because of its distance from the mechanics of action (Watkins, 2011). When faced with difficulties and negative events, concrete processing will be adaptive relative to abstract processing because it will result in (1) improved self-regulation focused on the immediate demands of the situation rather than its evaluative implications; (2) reduced negative overgeneralizations to emotional events, which contribute to increased emotional reactivity and vulnerability to depression; and (3) more effective problem solving.

Consistent with this theory, experimental studies have robustly found that abstract rumination causes negative consequences relative to concrete rumination, at least when focused on negative self-relevant information. As noted earlier, the standardized rumination induction was adapted into two variants that each retained the key original element of focus on self and mood, but with distinct instructions to induce concrete ("focus attention on the experience of") versus abstract ("think about the causes, meanings, and consequences of") processing. In depressed patients, compared with abstract rumination, concrete rumination reduced recall of overgeneral autobiographical memories (Watkins & Teasdale, 2001; 2004), reduced negative global self-judgments (Rimes & Watkins, 2005), improved recovery from prior negative events (Watkins, 2004), and improved social problem solving (Watkins & Moulds, 2005). Providing a conceptual replication to this finding, prompting abstract rumination (questions such as "Why did this problem happen?") impaired social problem solving in a recovered depressed group, who performed as well as never-depressed participants in a no-prompt control condition, whereas prompting concrete rumination ("How are you deciding what to do next?") ameliorated the problem-solving deficit found in currently depressed patients (Watkins & Baracaia, 2002).

Finally, repeated training to think in a concrete mode reduced subsequent emotional reactivity to analog loss and trauma events, relevant to training in an abstract mode (Watkins et al., 2008). Participants imagined thirty emotional scenarios (e.g., argument with best friend) in one or another mode of processing as training before a stressful anagram test. In the abstract training condition, for one minute per each scenario, participants were asked to "think about why it happened and to analyze the causes, meanings, and implications of this event." In the concrete training condition, for one minute per each scenario, participants were asked to "focus on how it happened and to imagine in your mind as vividly and as concretely as possible a 'movie' of how this event unfolded." The training mode was found to causally influence despondency after failure (Watkins et al., 2008).

Together, these studies provide convergent evidence that an abstract processing mode contributes to the negative consequences of rumination, relative to a more concrete mode. Because such abstract processing is characteristic of naturally occurring depressive rumination, it may be an important component of the negative effects of depressive rumination.

Because processing mode is observed to influence the consequences of rumination, a logical question is whether specificity of autobiographical memory retrieval also influences rumination. After all, overgeneral autobiographical memories are characterized by an abstract, generic form of mental representation, whereas specific autobiographical memories are characterized by recall of contextual and sensory-perceptual detail, necessarily involving more concrete representations. Thus, it is hypothesized that recall of categoric memories may causally contribute to rumination. To test this hypothesis, Raes et al. (2006) manipulated undergraduate students to recall either specific or overgeneral memories and then counted the number of sentences unscrambled with a ruminative meaning on a scrambled sentence task. In the specific retrieval style manipulation, participants were given word cues and asked to recall specific memories to the cues, whereas in the overgeneral retrieval style manipulation, participants were asked to recall categoric memories to the same cues. In high trait ruminators, relative to the specific retrieval condition, the overgeneral retrieval resulted in more ruminative completions. There was no effect of the memory manipulation in low trait ruminators. This study provides preliminary evidence that specificity of memory recall may influence ruminative processing, although it needs to be interpreted cautiously because the scrambled sentence task provided only an indirect measure of rumination and did not directly assess ongoing ruminative thinking.

Targeting abstract processing reduces rumination and depression

The experimental research reviewed above suggests that repeated and systematic targeting of abstract processing may be an efficacious intervention to reduce rumination and, thereby, treat depression. Consistent with the hypothesized causal relationship between processing mode and individual differences in rumination, a proof-of-principle randomized controlled treatment trial found that training stable dysphoric individuals (Beck Depression Inventory > 16 on two occasions separated by two weeks) to be more concrete when faced with difficulties reduced depression, anxiety, and rumination relative to a no-treatment control and a bogus training control (Watkins et al., 2009). The concreteness training involved repeated practice via direct instructions and guiding questions at (1) focusing on details in the moment (e.g., participants were asked to focus on and describe what they could see, hear, feel), (2) noticing what is specific and distinctive about the context of the event, (3) noticing the process of how events and behaviors unfolded (e.g., "imagine a movie of how events unfolded"), and (4) generating detailed step-by-step plans of how to proceed from here. Participants practiced audio-recorded exercises on a CD daily for one week.

In a follow-up phase II randomized controlled trial, concreteness training adapted as a facilitated self-help treatment was found to be superior to treatment-as-usual in reducing rumination, worry, and depression in 121 patients with major depression recruited in primary care (Watkins et al., 2012). The concreteness training involved one face-to-face session (ninety minutes), three thirty-minute phone sessions over six weeks, and daily practice on CD audio-recorded exercises. Patients were randomly allocated to treatment-as-usual as provided by general practitioners, concreteness training, and relaxation training, which was matched with concreteness training for structure, rationale, therapist contact, identification of warning signs, and active practice of CD exercises (in this case, progressive relaxation). There was no significant difference between relaxation training and concreteness training on reduction in depression, suggesting that identifying warning signs for rumination and practice at adopting an alternative coping strategy may be effective for reducing depression. Nonetheless, concreteness training outperformed relaxation training in reducing rumination and reducing negative attributional style.

One treatment designed specifically to target rumination is rumination-focused cognitive behavioral therapy (RFCBT). RFCBT is a manualized CBT treatment, consisting of up to twelve individual sessions scheduled weekly or fortnightly. RFCBT is theoretically informed by the research indicating that there are distinct constructive and unconstructive processing modes and is focused on inculcating the concrete processing mode in patients. Patients are coached to shift from the unconstructive abstract processing mode to the constructive concrete processing mode through the use of functional analysis, experiential/imagery exercises, and behavioral experiments. Functional analysis involves the close observation of behavior to establish the potential relationships between stimuli and responses within the principles of operant conditioning. Practically, it involves operationalizing a target behavior to be increased or decreased, and then identifying the antecedents and triggers, and the consequences of the behavior, to determine potential variables to manipulate in order to change the behavior. RFCBT incorporates the functional-analytic and contextual principles and techniques of behavioral activation (BA; Martell et al., 2001) but explicitly and exclusively focuses on rumination (for further details, see Watkins, 2009; Watkins et al., 2007;). RFCBT adopts the view that rumination is a form of avoidance and uses functional analysis to facilitate more helpful approach behaviors. RFBCT also uses functional analysis to help patients realize that their rumination about negative self-experience can be helpful or unhelpful and to coach them in how to shift to a more helpful style of thinking. In addition, patients use directed imagery to recreate previous mental states when a more helpful thinking style was active, such as memories of being completely absorbed in an activity (e.g., "flow" or "peak" experiences) and experiences of increased compassion, which act directly counter to rumination (for more information on therapeutic interventions related to imagery, see Clark and colleagues, Chapter 7). RFCBT significantly reduced rumination and depression in a multiple baseline case series of patients with residual depression (Watkins et al., 2007) and significantly outperformed treatment-as-usual (continuation antidepressants) in reducing rumination and depression in a phase II randomized controlled trial (Watkins et al., 2011). Moreover, over the six months of the trial, RFCBT significantly reduced relapse relative to treatment-as-usual.

In parallel to these treatment developments, there is an emergent line of research examining whether increasing the specificity of autobiographical memory retrieval can itself be therapeutic. Because overgeneral autobiographical memory retrieval has been implicated in increased vulnerability to depression, and because generic mental representations influence emotional reactivity, it is a plausible hypothesis that increasing the specificity of memory recall may reduce depression and rumination. Novel treatment approaches that target the specificity of autobiographical memories provide a proof-of-principle test of the hypothesis that generic processing contributes to the maintenance of rumination and depression. Two forms of memory training have recently been evaluated: Competitive Memory Training (COMET; Ekkers et al., 2011) and Memory Specificity Training (MEST; Raes et al., 2009).

In COMET, patients repeatedly practice generating specific counter-ruminative memories to situations in which rumination may occur, typically memories of times when the individual let go of concerns or was indifferent to difficult events. The recall of these specific memories is made as vivid as possible by utilizing posture, facial expression, and self-verbalizations to enhance imagery of these events. The COMET approach has parallels with both concreteness training, by encouraging patients to repeatedly practice an alternative response to warning signs for rumination, and with RFCBT, by involving a focus on generating experiential states that are counter to rumination. In a pilot randomized controlled trial, ninety-five older patients with major depression (aged sixty-five years or older) were randomized to seven weeks of COMET plus treatment-as-usual (TAU) versus TAU only. COMET plus TAU showed a significant improvement in depression and rumination compared with TAU alone. This provides initial evidence of this approach but leaves unresolved the mechanism of the effect – without an active treatment control, we do not know whether these effects are due to nonspecific treatment effects or which element of COMET was efficacious.

Memory Specificity Training (MEST; Raes et al., 2009) involves four one-hour-long group sessions in which patients are educated about depression and memory specificity, and then practice recalling positive and neutral specific memories that include contextual and sensory-perceptual details, before moving on to recall of specific memories to negative cues. In a case series of ten patients with depression, Raes et al. (2009) found that the training improved specificity on the autobiographical memory test and reduced rumination and symptoms of depression. However, this was an uncontrolled design, and a randomized controlled trial of MEST is required.

Together, these studies provide evidence that training patients with depression to become more concrete and more specific is therapeutic, whether this training is focused on how events are represented (concreteness training) or on the recall of memories to cues (MEST). A key next step is to examine whether these interventions are all acting on common elements (e.g., Does concreteness training improve specificity of memory retrieval? Does MEST improve problem solving?) and identify the active mediators of these interventions.

Nonetheless, these results are consistent with the hypothesis that increasing specificity may be an important mechanism in effective psychological interventions for depression (Watkins, 2009). Increasing specificity and concreteness is an element within traditional CBT for depression, a major focus within RFCBT, and the primary target of concreteness training and MEST. Moreover, as noted earlier, overgeneralization is an important cognitive bias in depression, and these interventions may reduce this bias. There is also evidence implicating increased concreteness and specificity within effective treatment: concrete methods (e.g., asking for specific examples) predict subsequent symptom reduction when assessed early in CBT, whereas abstract methods (e.g., discussion of meaning) do not (DeRubeis & Feeley, 1990; Feeley et al, 1999). Furthermore, improvement in situational analysis, which involves specific and detailed descriptions of problems and their context, in the first six weeks of CBT predicted reduced depression at the end of treatment (Manber et al., 2003).

Common theoretical constructs between rumination and overgeneral memory: discrepancy and hierarchy

The evidence reviewed above suggests that manipulating rumination can influence overgeneral memory but also that manipulating specificity of autobiographical memory can influence rumination. The unifying factor across this bidirectional causal relationship appears to be the adoption of more abstract processing, which drives both increased overgeneral autobiographical memory and the negative consequences of rumination.

I propose that this common role of abstraction in both memory retrieval and rumination emerges from common theoretical properties implicated in models of both rumination and overgeneral memory, namely, iterative processing across multiple levels of information organized in a hierarchy. Control theory accounts of rumination hypothesize that rumination is triggered and maintained by unresolved goal discrepancies (Martin & Tesser, 1996; Watkins, 2008). Rumination is proposed to continue in an iterative fashion until this goal discrepancy is resolved, through either attaining or abandoning the goal. Similarly, the generative search involved in voluntary autobiographical memory retrieval is hypothesized to involve the matching of information retrieved against search criteria. This search continues iteratively until the search criteria are satisfied, that is, there is no discrepancy between information generated and search criteria. Consistent with this account, there is evidence that overgeneral memory recall is more likely to occur when participants are asked to retrieve specific memories to positive cue words after a self-discrepancy induction (rating positive trait words for the extent that the participant currently possessed this quality versus the extent that the participant ideally wants to possess this quality; Raes et al., 2012; Schoofs et al., 2012).

Furthermore, the voluntary recall of overgeneral autobiographical memory is located within a hierarchical model of generative memory search (Williams et al., 2007), in which lifetime period information leads down to generic information, which in turn leads to specific (sensory-perceptual) information. Likewise, recent theoretical accounts of rumination have located it within a hierarchical model of goal and action identification (Watkins, 2008; 2011). Within these accounts, goals and actions are organized hierarchically: abstract levels in the hierarchy represent the ends and meaning of a goal or action (e.g., "why" a goal or action is sought or performed), whereas more subordinate concrete levels represent the detail and means of "how" the superordinate goal or action is enacted (e.g., programs and sequences of actions). Within this hierarchical organization, pursuit toward abstract goals occurs by specifying reference values at the next lower level of abstraction, all the way down to the concrete representations required to specify the actual behaviors needed to progress toward the goal. For example, one classification of levels within control theory approaches (Carver & Scheier, 1990) proposes that the reference values at the most abstract levels may represent a global sense of idealized self (i.e., a decontextualized, superordinate meaning capturing the essence of the self), which in turn sets the broad principles that organize goals and behavioral standards across multiple situations (e.g., to be an honest person), whereas the reference values at the more concrete levels represent the specific actions and behavioral programs and sequences necessary to implement the principles in a particular situation (e.g., telling the truth to a friend, i.e., more contextualized, specific details of how to do the action).

Control theory accounts further propose that a particular level in the hierarchy may be functionally and operationally prepotent and superordinate at any moment in time, reflecting whether the individual is focusing attention and awareness on a more abstract or concrete level, and thereby representing reference values (goals) and perceptual signals (from the environment) in a more abstract or concrete manner (i.e., this corresponds to the adoption of a concrete or abstract processing mode).

Furthermore, this theoretical model can account for the prior experimental findings that processing difficult and stressful situations at the abstract level may be problematic. Depending on context, a level of control that is too abstract or too concrete or that fails to link abstract levels to concrete levels is hypothesized to be detrimental (Carver & Scheier, 1998). Processing at an abstract level can be problematic because it can (1) make the relevant concern more personally important, resulting in greater emotional distress when the goal is not achieved and making it harder to abandon the unresolved goal, and (2) provide less specification of alternatives and concrete steps to proceed, impairing problem solving.

Thus, both overgeneral memory retrieval and rumination share the tendency, perhaps functionally reinforced, to preferentially adopt a more abstract level of processing when moving through their respective hierarchies of information. This can be conceptualized as both depressive rumination and overgeneral memory retrieval adopting abstract construals focused on decontextualized mental representations that convey the essential gist and meaning of events, goals, and actions. It is important to recognize that this observed relationship between overgeneral memory retrieval and rumination is focused on voluntary memory retrieval, such as studied with the Autobiographical Memory Test. Although there is growing research into involuntary autobiographical memories (for further information, see Moulds and Krans, Chapter 8; Berntsen, Chapter 9), the relationship between rumination and involuntary autobiographical memory is largely unexamined, with the exception of Watson et al. (2013), who found rumination correlated with reduced specificity of both voluntary and involuntary memories in depressed patients. Determining the causal relationship between rumination and involuntary autobiographical memory is a priority for future research.

Conclusion

In sum, the evidence reviewed in this chapter suggests that rumination causally influences the retrieval of overgeneral autobiographical memories and overgeneral thinking, but also that more abstract and generic processing, including the retrieval of categoric memories, causally influences the consequences and frequency of rumination. Moreover, interventions that target abstract processing, including those that increase specificity of autobiographical memory, reduce rumination and depression. There is thus robust evidence indicating a bidirectional relationship between rumination and overgeneral autobiographical memory, with each mutually reinforcing the other. Moreover, it is proposed that this mutual reinforcement occurs as a function of increased abstraction of representations being shared across depressive rumination and overgeneral autobiographical memory, reflecting iterative processing through hierarchical models being common to theoretical accounts of both phenomena.

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